MYGA Customer Analysis

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April 3, 2016

Overview

During both MYGA specials, iPipeline retained all answers to suitability questions answered by clients. This gives us a very interesting look into the demographics of these policyholders. The analysis pulls from file with over 600,000 records of all apps submitted through the Affirm platform. Some of the records include cases that may not have been placed and may possibly include some test cases. We will progressively clean the file as we receive better information from Dell and FGL systems.

Sales Volume

This file includes records of 5,970 applications that represented $532.5 million in premiums. Each sale averaged just under $90,000. The median case size was $50,000. Though we saw a wide range of case sizes, we did see four meaningful clusters at $25,000, $50,000, $100,000 and $150,000.

x

Cases

5970

Total_Premium

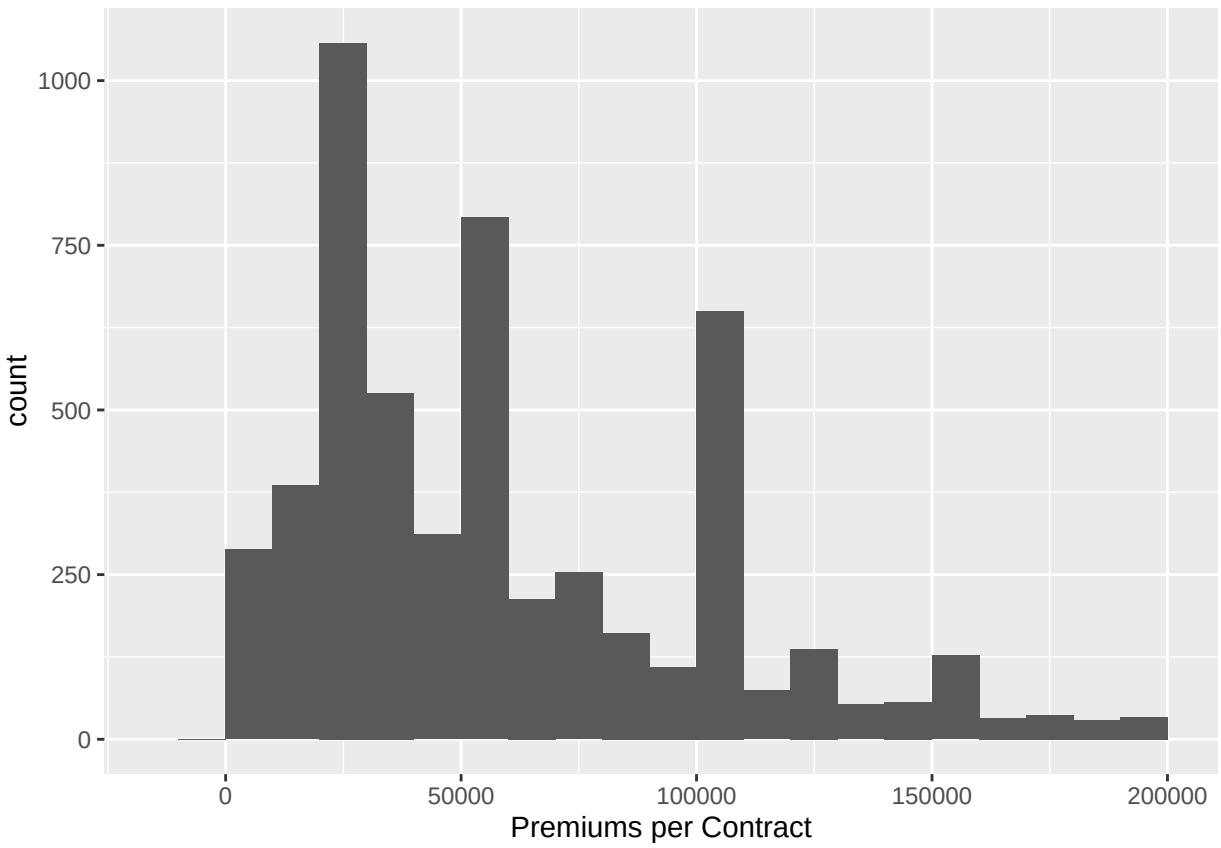
532508630

Avg_Premium

89197

Med_Premium

50000



Net Worth of Consumers

A few customers reported very large networths and pushed the average networth to $1.5 million. We think the median networth of slightly less than $1 million - $995,000 - presents a truer picture. Roughly half the assets are liquid and the other half is illiquid.

x

mean_assets

1486247

mean_liq_assets

759464

mean_non_liq_assets

725579

median_assets

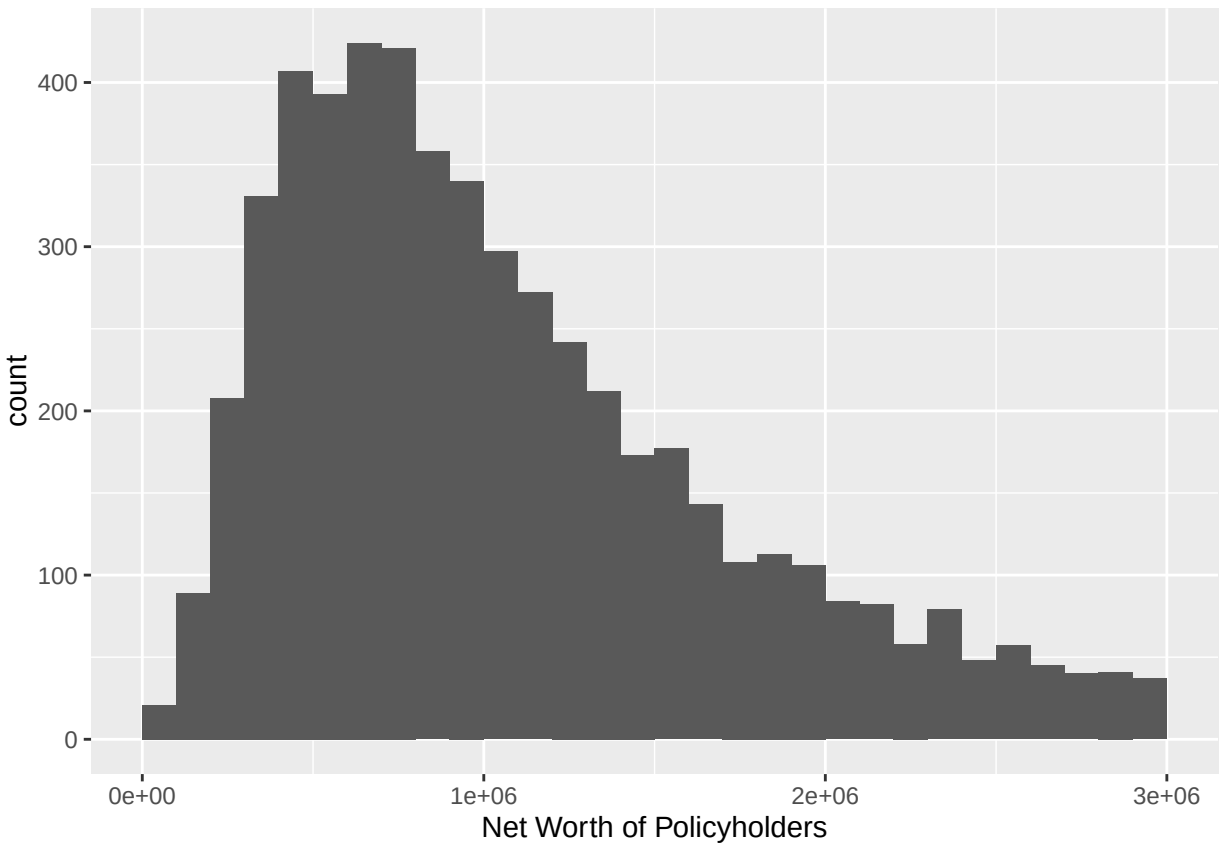
995000

median_liq_assets

465000

median_non_liq_assets

431500



Source of Net Worth

The largest source of net worth reported by clients was their home, their qualified retirement plan, and their checking or money market account, in that order. The vast majority of the money used to purchase the annuity came from liquid sources.

Avg_Liq_Src

Avg_Non_Liq_Src

1

0.93

0.07

Liquid Assets

Most of the liquid net worth comes first from a qualified plan (i.e. IRA, 401(k)) and then from a checking or money market account. Some large outliers in all categories distort the average amount.

```
##              [,1]      [,2]
## Value        "Average" "Median"
## Cash         "54080"   " 2000"
## Checking     "145410"  " 68000"
## CDs          "76034"   "    0"
## IRAs         "202732"  " 76000"
## Stocks_Bonds "96529"   "    0"
```

```
## Mutual_Funds "66449"   "    0"
## Annuities    "87883"   "    0"
## Life_Ins     "14453"   "    0"
## Other        "14812"   "    0"
```

Non-Liquid Assets

The house is typically the major non-liquid asset reported by clients.

```
##                [,1]      [,2]
## Value          "Average" "Median"
## Val_of_Home    "296288"  "230000"
## Other_RE       "200248"  "     0"
## IRA_under      "52937"   "    0"
## Class_B_MF     "2406"    "   0"
## Gold           "11468"   "    0"
## Ann_with_Pen   "134562"  "     0"
## Life_with_Pen  "4292"    "   0"
## Other_Non_Liq  "23378"   "    0"
```

Home Values

The average value of the home was close to $300,000. The median was exactly $230,000.

x

minimum

0

mean

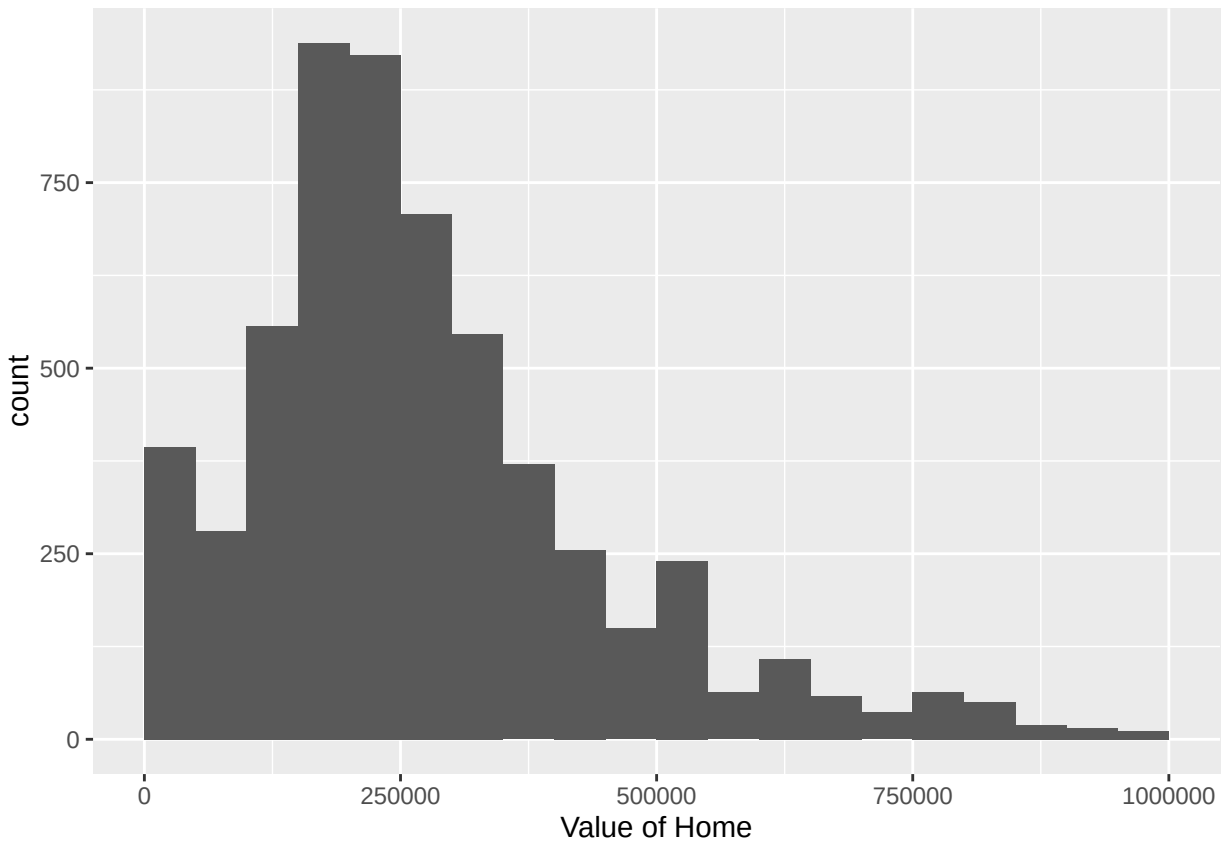
296288

median

230000

maximum

6000000



Age

We received the age of owners for the latest MYGA program. The average age of our client is 68 and the oldest was 90.

Avg_age

Med_age

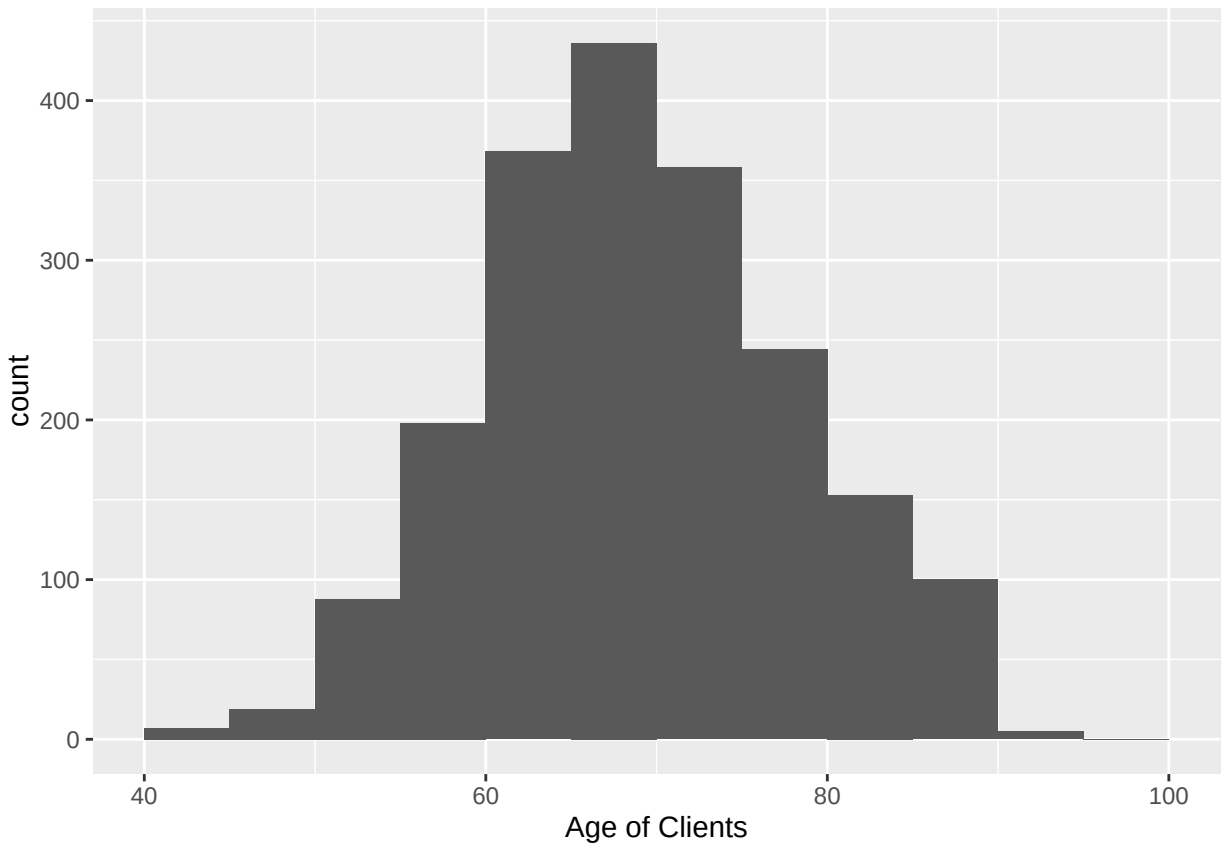
Max_age

1

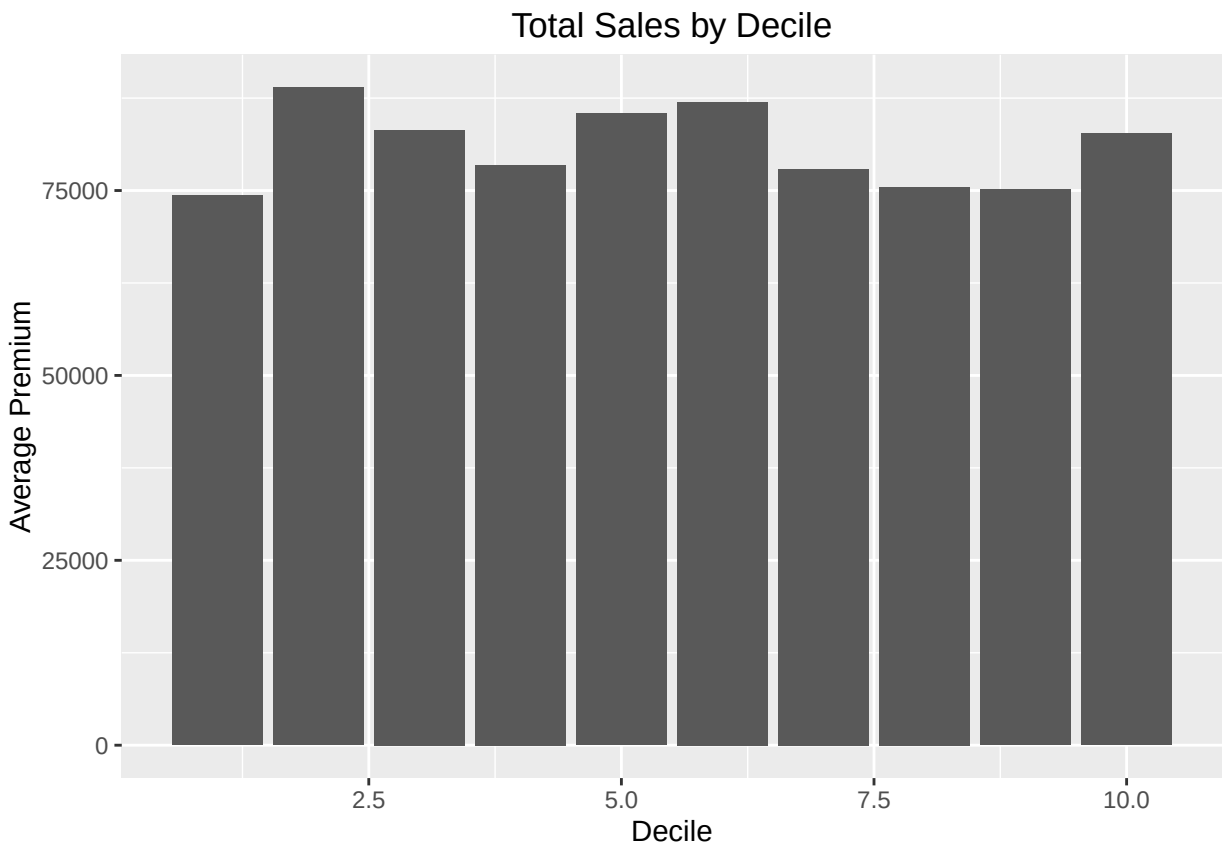
68.3

68

90



We saw little difference in average case size in any age band.



Decile_Age

Min_age

Max_age

Avg_case

Total_sales

Tot_Cases

1

1

2

57

74328

14791242

199

2

2

57

61

88920

17695046

199

3

3

61

63

83200

16556726

199

4

4

63

66

78430

15529202

198

5

5

66

68

85402

16995044

199

6

6

68

70

86987

17310328

199

7

7

71

73

77899

15424065

198

8

8

73

77

75490

15022453

199

9

9

77

81

75107

14946350

199

10

10

81

90

82689

16372391

198

Discretionary Income

We ask applicants for their monthly income and monthly expenses. The difference or median discretionary income to spend is $2,500.

x

minimum_discr

-55000

mean_discr

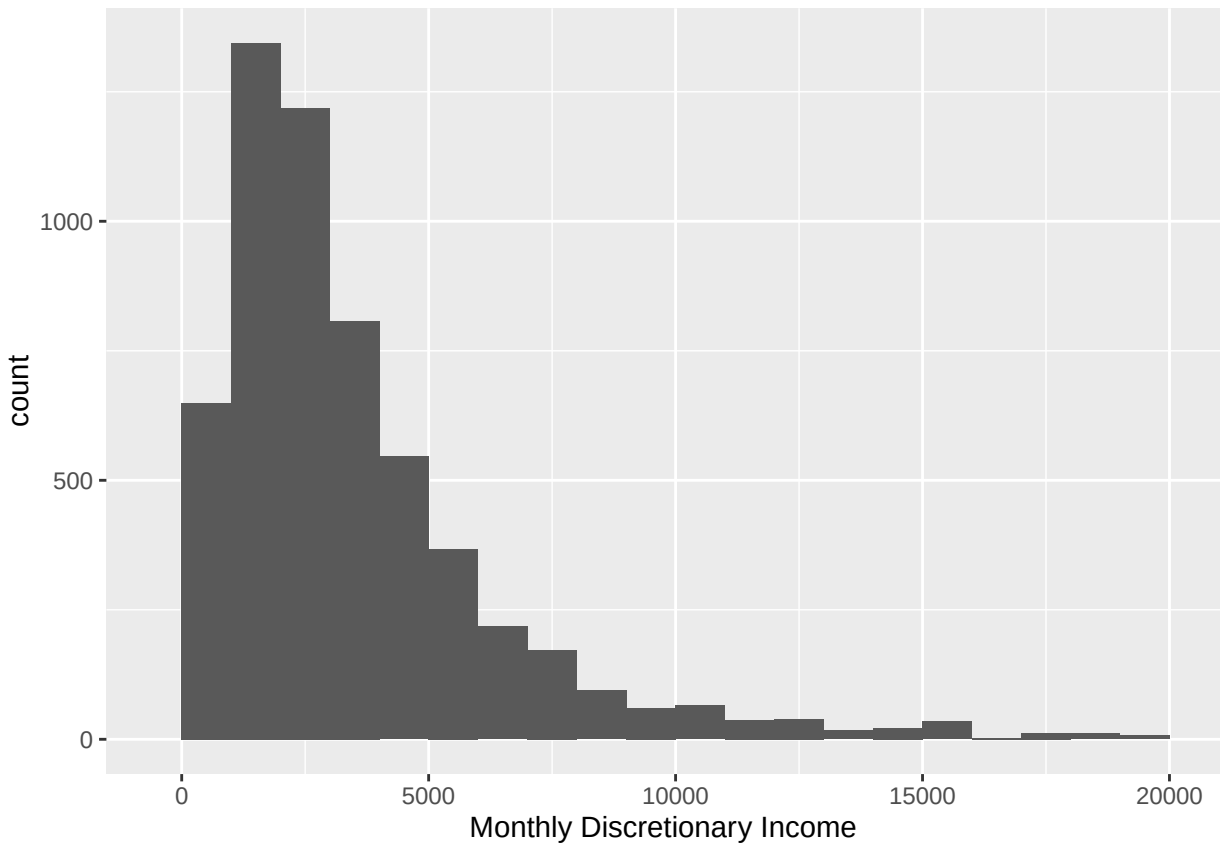
5410

median_discr

2500

maximum_discr

983000



Work Status

Close to 70% owners as well as any joint owners reported they were "retired." Slightly more than 30% indicated they were employed.

Owner

ct

status

n

pct

1

1.00

Retired

4011

0.67

2

2.00

Employed

1843

0.31

3

3.00

Unemployed

45

0.01

4

4.00

Other

71

0.01

Joint Owner

ct

status

n

pct

1

1.00

Retired

443

0.68

2

2.00

Employed

178

0.27

3

3.00

Unemployed

11

0.02

4

4.00

Other

16

0.02

Gender

In the last wave of sales, 45% of customers were women, 55% were men. Cases sold to men were slightly higher larger than those sold to women.

Insured.Gender

Avg_age

Avg_case

Total_sales

Pct_sales

Tot_Cases

Pct_Cases

Mean_Home

Mean_Assets

1

F

68

74551

71345095

0.44

957

0.48

265675

1346541

2

M

68

86697

89297750

0.56

1030

0.52

289100

1588697

Impact of Accepting 1035s

We launched our second online MYGA program begining November 30, 2015 as cash-only offer. On January 20, 2016, we began accepting 1035 exchanges. The change in process did not appear to impact the size of cases. However, the daily sales level picked up substantially after the announcement. The rate hike from 3% to 3.15% on 2/26 created the very large spike in March.

Time

Cases

Avg_Premium

Med_Premium

Avg_Assets

Med_Assets

1

Post-1035s

1467

81389

50000

1401544

1000000

2

Pre-1035s

520

79318

50000

1671025

991000

That said, accepting 1035 exchanges allowed us to increase the size of cases by approximately 20%.

Is.1035

Cases

Total_Premium

Avg_Premium

Med_Premium

Avg_Liq_Src

Avg_Non_Liq_Src

1

No

1425

109218651

76645

50000

0.95

0.05

2

Yes

562

51424195

91502

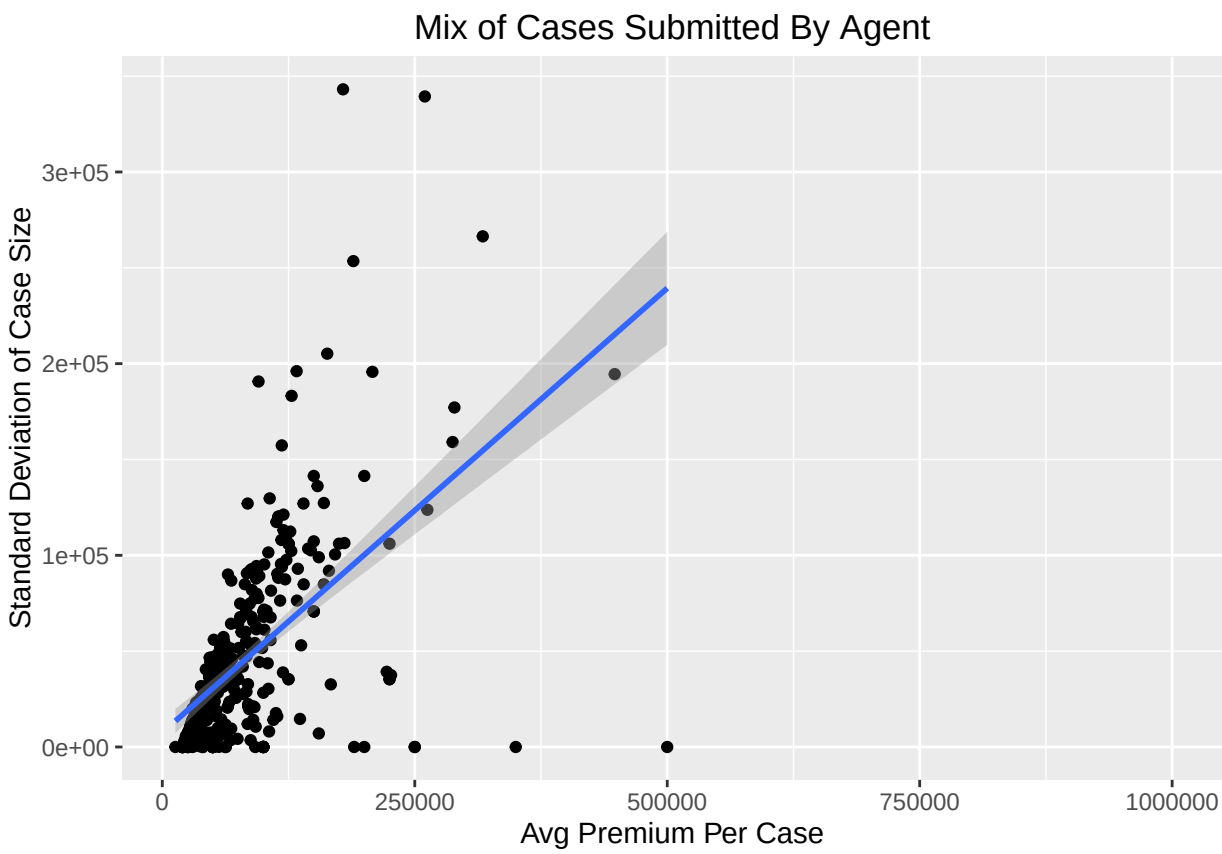
60071

0.88

0.12

Role of the Agent

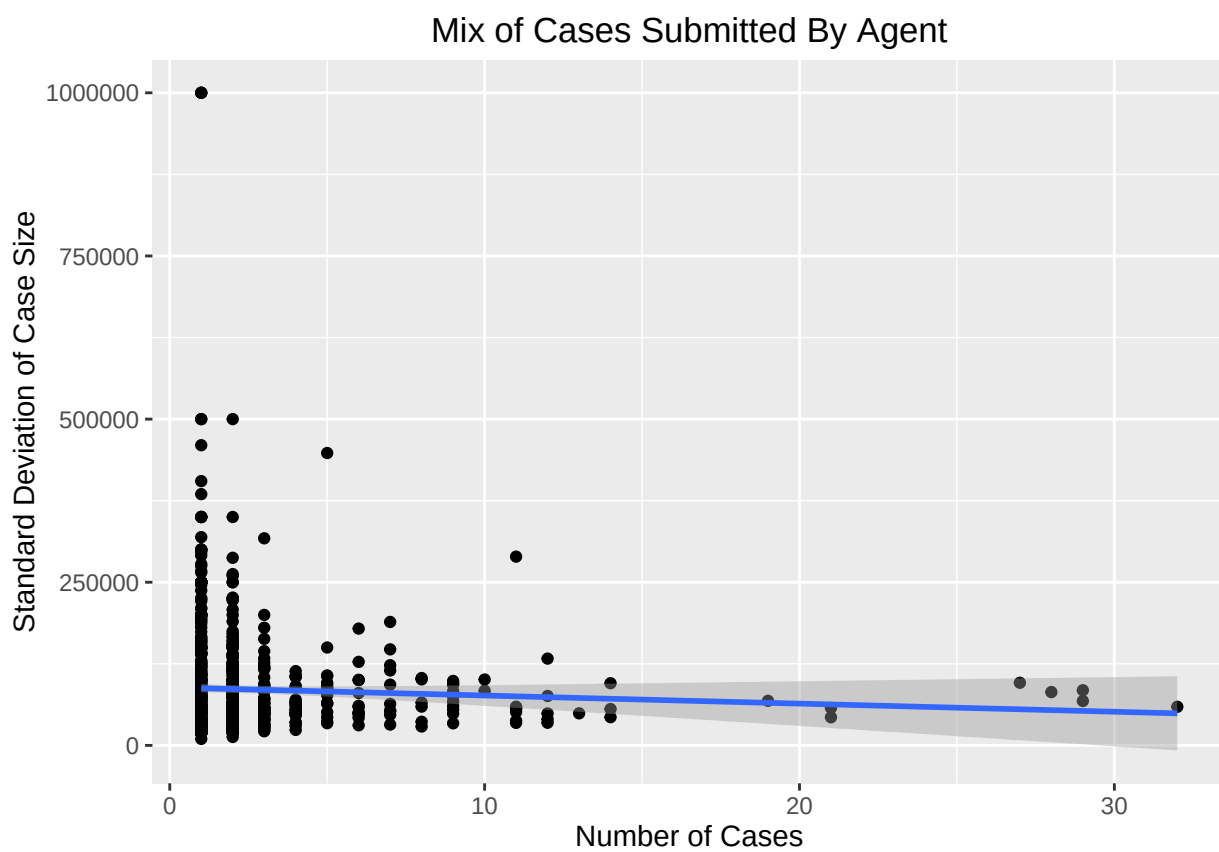
It appears that agents generally sold the same size of case during the last special. No segment of agents seems to have consistently sold larger cases than the overall population.




```
##
## Call:
## lm(formula = agents$St_dev ~ agents$Avg_case_size)
```

```
##
## Coefficients:
##          (Intercept)  agents$Avg_case_size
##            7509.7884                0.4637
```

```
##
## Call:
## lm(formula = agents$St_dev ~ agents$Avg_case_size)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -239363  -16784   -3799   14884  252619
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          7.510e+03  3.584e+03   2.095   0.0368 *
## agents$Avg_case_size 4.637e-01  3.545e-02  13.080   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 40790 on 360 degrees of freedom
##   (517 observations deleted due to missingness)
## Multiple R-squared:  0.3221, Adjusted R-squared:  0.3203
## F-statistic: 171.1 on 1 and 360 DF,  p-value: < 2.2e-16
```




```
##
```

```
## Call:
## lm(formula = agents$St_dev ~ agents$Cases)
##
## Coefficients:
##  (Intercept)  agents$Cases
##        37694          1819
```

```
##
## Call:
## lm(formula = agents$St_dev ~ agents$Cases)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -48607 -34532 -13513  15066 298079
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   37694.1     3581.7  10.524  < 2e-16 ***
## agents$Cases   1818.8      613.7   2.964  0.00324 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 48950 on 360 degrees of freedom
##   (517 observations deleted due to missingness)
## Multiple R-squared:  0.02382,    Adjusted R-squared:  0.02111
## F-statistic: 8.785 on 1 and 360 DF,  p-value: 0.00324
```

Concentration of Business

Distribution by Agent Sales Volume

Large cases are greater than $150,000

Decile_Sales

Avg_case_size

Med_case_size

Total_sales

Avg_cases

Avg_lg_cases

Tot_Cases

St_dev

1

1

21571

20000

1898263

1.0

0.0

88

2

2

30480

30000

2775246

1.1

0.0

93

0

3

3

43399

50000

4174846

1.2

0.0

104

1346

4

4

51853

51000

5372056

1.3

0.0

115

10118

5

5

75447

90000

8180371

1.5

0.0

130

9405

6

6

85340

100000

9296562

1.4

0.0

127

24860

7

7

94761

77300

13212845

2.0

0.1

172

31616

8

8

126669

108836

18607332

2.2

0.5

198

45595

9

9

160590

139910

26972641

2.9

0.7

255

65688

10

10

171612

80000

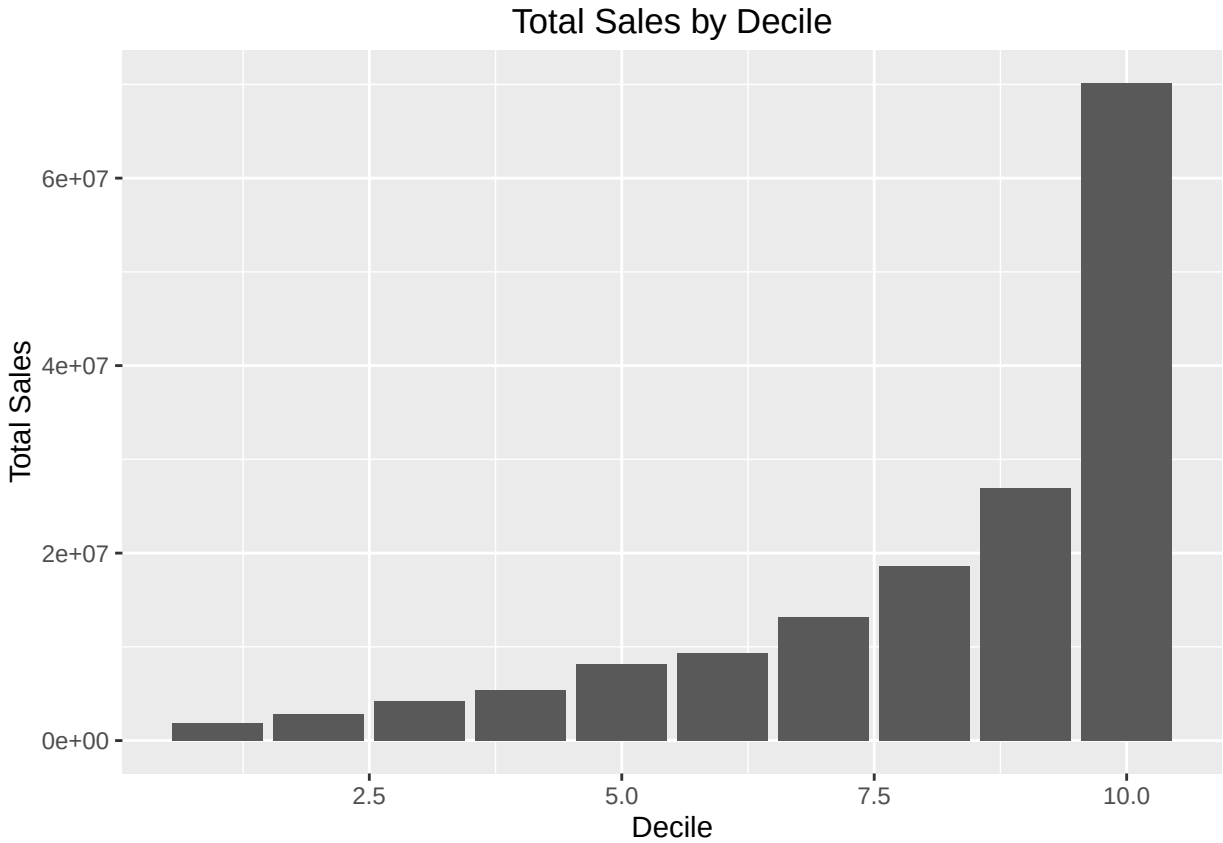
70152683

8.1

1.2

705

80747



Distribution by Agent Case Volume

Large cases are greater than $150,000

Decile_Cases

Total_cases

Avg_cases

Avg_lg_cases

Avg_case_size

Med_case_size

Total_sales

St_dev

1

1

88

1.0

0.1

97821

59926

8608236

2

2

88

1.0

0.2

110034

75000

9682953

3

3

88

1.0

0.1

78203

50000

6881882

4

4

88

1.0

0.1

74257

50000

6534616

5

5

88

1.0

0.1

86396

50000

7602888

6

6

99

1.1

0.1

88584

60500

8571050

27554

7

7

176

2.0

0.2

78883

69839

13883381

37523

8

8

180

2.0

0.4

97496

71354

17418982

40030

9

9

284

3.2

0.3

68794

50000

19399915

42038

10

10

808

9.3

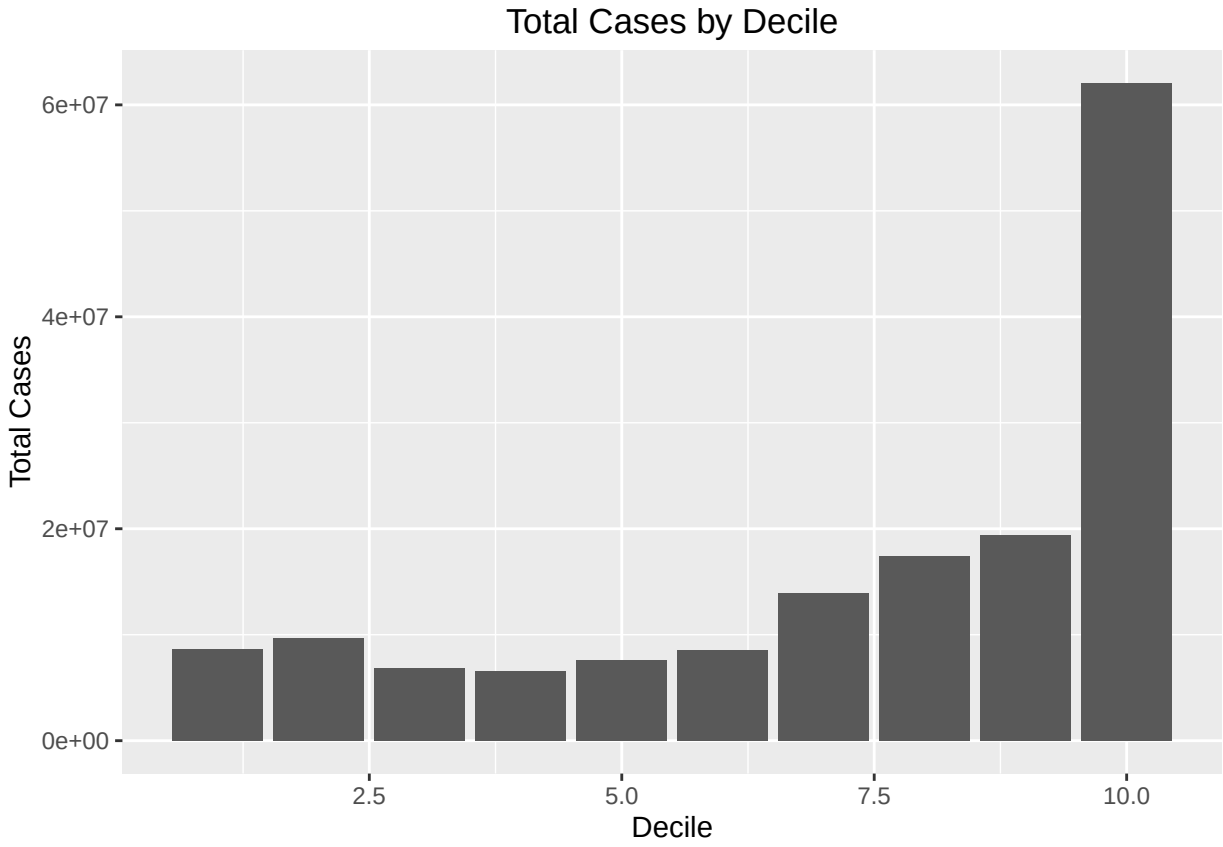
0.8

80217

50000

62058944

63124



Multivariate regression model

```
##
## Call:
## lm(formula = Purchase ~ Insured.Gender + Is.1035 + Checking,
##     data = model_purchase)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -383883  -42173  -23836   18289  933949
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     5.384e+04  3.165e+03  17.012  < 2e-16 ***
## Insured.GenderM 1.102e+04  3.804e+03   2.896  0.00382 **
## Is.1035Yes      2.292e+04  4.255e+03   5.386 8.06e-08 ***
## Checking        9.226e-02  6.042e-03  15.270  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 84690 on 1983 degrees of freedom
## Multiple R-squared:  0.114,  Adjusted R-squared:  0.1127
## F-statistic: 85.08 on 3 and 1983 DF,  p-value: < 2.2e-16
```